Lecture 8

Tempered distributions and Green's functions on the line

Objectives. *A point source is described by its action on smooth functions rather than by pointwise values. We define that action, extend differentiation and the Fourier transform to it, and prove the boundary-value formula for a simple pole. We then solve two differential equations with a point source.*

By the end of this lecture you will be able to:

- Bound a tempered distribution by finitely many Schwartz seminorms.
- Compute distributional derivatives and polynomial multipliers.
- Derive the Plemelj signs and the unitary Fourier transform table.
- Verify fundamental solutions and solve the corresponding forced equations.

8.1 Continuous linear functionals on Schwartz space

Recall from Lecture 7 that $\mathcal{S}(\mathbb{R})$ consists of smooth functions with finite seminorms

$$\|\varphi\|_{a,b} = \sup_{x\in\mathbb{R}} \left| x^a \varphi^{(b)}(x) \right|, \qquad a, b \in \mathbb{N}_0.$$

A basic *neighbourhood of zero* imposes finitely many bounds $\|\varphi\|_{a,b} < r_{a,b}$, with all $r_{a,b} > 0$. Neighbourhoods of another function are translates of these sets. A linear functional T is *continuous* if, for every $\varepsilon > 0$, some such zero neighbourhood makes $|T[\varphi]| < \varepsilon$. This is continuity at zero; linearity gives continuity everywhere. A sequence converges in $\mathcal{S}$ when every seminorm of its difference from the limit tends to zero. A *tempered distribution* is a continuous complex-linear functional $T : \mathcal{S}(\mathbb{R}) \to \mathbb{C}$. The space of these functionals is $\mathcal{S}'(\mathbb{R})$. The prime denotes a dual space, not a derivative. We write its value as $T[\varphi]$. The pairing is bilinear: there is no complex conjugate in either argument. Convergence in $\mathcal{S}'$ means convergence of the values on every fixed $\varphi \in \mathcal{S}$.

Proposition 8.1 (Seminorm bound)**.** *A linear functional T is continuous exactly when there are $C > 0$ and finitely many pairs (a, b) such that*

$$|T[\varphi]| \le C \sum_{(a,b)\ \text{in the finite set}} \|\varphi\|_{a,b} \qquad (\varphi \in \mathcal{S}).$$

Proof. Such a bound gives continuity directly. Conversely, continuity at zero gives a neighbourhood, defined by finitely many conditions $\|\varphi\|_{a,b} < r$, after taking the minimum of their radii, on which $|T[\varphi]| < 1$. Let $q(\varphi)$ be the sum of those seminorms. If $q(\varphi) > 0$, apply the

neighbourhood bound to $r\varphi/(2q(\varphi))$ and use linearity. If $q(\varphi) = 0$, every scalar multiple lies in that neighbourhood, so linearity forces $T[\varphi] = 0$. The required constant is $2/r$. ∎

A measurable function with $|f(x)| \le C(1+|x|)^N$ a.e., for some integer $N \ge 0$, defines a distribution by $f[\varphi] = \int_{\mathbb{R}} f(x)\varphi(x)\,dx$. Indeed, the integral is bounded by a constant times $\sup_x (1+|x|)^{N+2}|\varphi(x)|$, since $\int (1+|x|)^{-2}\,dx < \infty$. Expanding the integer power bounds this supremum by finitely many of the stated seminorms. Constant functions, polynomials, the *Heaviside function* $\theta = \mathbb{1}_{(0,\infty)}$, and the sign function $\operatorname{sgn} x$ therefore define tempered distributions. The value assigned at a single point does not change their integrals.

Every $f \in L^1(\mathbb{R})$ also defines a tempered distribution, since $|f[\varphi]| \le \|f\|_1 \|\varphi\|_{0,0}$. For $f \in L^2(\mathbb{R})$, Hölder's inequality (Theorem 5.1) gives $|f[\varphi]| \le \|f\|_2 \|\varphi\|_2$. Splitting at $|x| = 1$ bounds the test norm by

$$\|\varphi\|_2^2 \le 2\|\varphi\|_{0,0}^2 + \frac{2}{3}\|\varphi\|_{2,0}^2 \le 2(\|\varphi\|_{0,0} + \|\varphi\|_{2,0})^2.$$

Here the tail uses $|\varphi(x)| \le \|\varphi\|_{2,0}|x|^{-2}$ and $\int_{|x|>1} |x|^{-4}\,dx = 2/3$. Thus the L^2 pairing is controlled by the finite-seminorm bound

$$|f[\varphi]| \le \sqrt{2}\,\|f\|_2\,(\|\varphi\|_{0,0} + \|\varphi\|_{2,0}). \tag{8.1}$$

In particular, convergence in L^2 gives convergence on each Schwartz test.

The *Dirac distribution at* $a \in \mathbb{R}$ is $\delta_a[\varphi] = \varphi(a)$; write $\delta = \delta_0$. Its bound is $|\delta_a[\varphi]| \le \|\varphi\|_{0,0}$. The *principal value* distribution is defined by

$$\operatorname{PV}(1/x)[\varphi] = \lim_{\varepsilon \downarrow 0} \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\,dx.$$

To prove existence and continuity, write it as

$$\int_{|x|<1} \frac{\varphi(x)-\varphi(0)}{x}\,dx + \int_{|x|\ge 1} \frac{\varphi(x)}{x}\,dx. \tag{8.2}$$

The first integral is bounded by $2\|\varphi\|_{0,1}$, by the fundamental theorem of calculus. The second is bounded by $\|\varphi\|_{2,0}\int_{|x|\ge 1}|x|^{-3}\,dx = \|\varphi\|_{2,0}$. The same estimates justify the limit defining it.

The function e^x does not define a tempered distribution extending its integrals on compactly supported tests. Choose a nonzero nonnegative $j \in C_c^\infty(0,1)$ and put $\varphi_n(x) = e^{-n/2}j(x-n)$. Every Schwartz seminorm tends to zero, since it is bounded by a constant times $e^{-n/2}(n+1)^a$. But $\int e^x\varphi_n(x)\,dx = e^{n/2}\int_0^1 e^y j(y)\,dy$ tends to infinity. Continuity fails. The larger space $\mathcal{D}'(\mathbb{R})$ of distributions on compactly supported smooth tests accommodates this function; we do not use it here.

8.2 Derivatives and polynomial multipliers

For $T \in \mathcal{S}'$ and a polynomial p, define

$$T'[\varphi] = -T[\varphi'], \qquad (pT)[\varphi] = T[p\varphi].$$

Differentiation shifts the derivative index in a seminorm. The product rule bounds each seminorm of $p\varphi$ by finitely many seminorms of φ. Thus these operations preserve $\mathcal{S}'$. For

$f \in C^1(\mathbb{R})$, integration by parts gives agreement with its ordinary derivative when f and f' are polynomially bounded. Directly from the definitions,

$$\theta'[\varphi] = -\int_0^\infty \varphi'(x)\,\mathrm{d}x = \varphi(0), \qquad x\delta = 0, \qquad x\,\mathrm{PV}(1/x) = 1.$$

In particular, $\theta' = \delta$ and $\delta'[\varphi] = -\varphi'(0)$. The sign function satisfies $\operatorname{sgn} = 2\theta - 1$, hence $(\operatorname{sgn})' = 2\delta$. These identities hold for arbitrary tests, not only away from zero.

The locally integrable function $\ln|x|$ is tempered. Near zero, $\int_0^1 |\ln x|\,\mathrm{d}x = 1$ gives a bound by $\|\varphi\|_{0,0}$. At infinity its growth is bounded by a polynomial. Integration by parts off $(-\varepsilon, \varepsilon)$ gives

$$-\int_{|x|>\varepsilon} \ln|x|\ \varphi'(x)\,\mathrm{d}x = \int_{|x|>\varepsilon} \frac{\varphi(x)}{x}\,\mathrm{d}x + \ln\varepsilon\,[\varphi(\varepsilon) - \varphi(-\varepsilon)].$$

The last term tends to zero because the difference is bounded by $2\varepsilon\,\|\varphi\|_{0,1}$. The omitted integral near zero also tends to zero by local integrability. Therefore $(\ln|x|)' = \mathrm{PV}(1/x)$. Our operations define polynomial times distribution. They do not define products of arbitrary distributions, such as δ^2.

8.3 The Plemelj boundary-value formula

Theorem 8.2 (Plemelj lemma). *Let $\varphi \in C^1(\mathbb{R})$ and suppose $\int_{|x|\ge 1} |\varphi(x)/x|\,\mathrm{d}x < \infty$. Then the principal value exists and*

$$\lim_{\varepsilon\downarrow 0} \int_{\mathbb{R}} \frac{\varphi(x)}{x \mp \mathrm{i}\varepsilon}\,\mathrm{d}x = \mathrm{PV}\int_{\mathbb{R}} \frac{\varphi(x)}{x}\,\mathrm{d}x \ \pm \mathrm{i}\pi\varphi(0).$$

The notation $1/(x \mp \mathrm{i}0)$ denotes the limit in $\mathcal{S}'$ of $1/(x \mp \mathrm{i}\varepsilon)$ as $\varepsilon \downarrow 0$. Consequently $1/(x \mp \mathrm{i}0) = \mathrm{PV}(1/x) \pm \mathrm{i}\pi\delta$ in $\mathcal{S}'$.

Proof. Existence of the principal value follows from the subtraction in (8.2), with the given integrable tail. For the upper choice,

$$\frac{1}{x - \mathrm{i}\varepsilon} = \frac{x}{x^2 + \varepsilon^2} + \mathrm{i}\frac{\varepsilon}{x^2 + \varepsilon^2}.$$

On $|x| < 1$, subtract $\varphi(0)$ in the first integral. Its constant contribution vanishes by oddness. The remaining integrand is bounded by $\sup_{[-1,1]} |\varphi'|$, and tends to $(\varphi(x) - \varphi(0))/x$. On $|x| \ge 1$, use the dominator $|\varphi(x)/x|$. Dominated convergence (Theorem 3.5) proves convergence to the principal value.

For the second integral, its constant part on $(-1, 1)$ is $2\varphi(0)\arctan(1/\varepsilon)$, which tends to $\pi\varphi(0)$. The remainder on this interval tends to zero by the same theorem: its modulus is at most $\sup_{[-1,1]} |\varphi'|\ \varepsilon\,|x|\,/(x^2 + \varepsilon^2)$, which is bounded by half that supremum. For $0 < \varepsilon \le 1$, the tail is dominated by $|\varphi(x)/x|$ and tends pointwise to zero. Changing the sign of the imaginary part proves the lower choice. Every Schwartz function meets the hypotheses, giving the distributional statement. ∎

The notation i0 specifies a limit from one side. It is not a small number that can be cancelled algebraically. The two limits differ by $2\pi\mathrm{i}\delta$, although their values away from zero both tend to $1/x$.

8.4 The Fourier transform of distributions

The Fourier transform of a tempered distribution is defined by

$$(\mathcal{F}T)[\varphi] = T[\mathcal{F}\varphi].$$

The continuous map $\mathcal{F} : \mathcal{S} \to \mathcal{S}$ from Theorem 7.3 and the seminorm bound show that $\mathcal{F}T$ is tempered. Fubini (M4, Theorem 4.1) proves agreement with the integral transform for L^1 functions: the kernel $e^{-\mathrm{i}kx}$ is symmetric in x and k, so $\int \widehat{f}\varphi = \int f\widehat{\varphi}$. There is no conjugation in this identity. The inverse is defined using $\mathcal{F}^{-1}$ on tests. By Theorem 7.5, $\mathcal{F}^2T[\varphi] = T[\varphi(-\cdot)]$. Convergence on every test is preserved by $\mathcal{F}$.

Proposition 8.3. *For $f \in L^2(\mathbb{R})$, the distributional Fourier transform is the distribution represented by its Plancherel transform.*

Proof. By Theorem 5.8, choose $f_j \in \mathcal{S}$ tending to f in L^2. For each $\varphi \in \mathcal{S}$, Fubini on these integrable functions gives

$$\int_{\mathbb{R}} \widehat{f_j}(k)\varphi(k)\,\mathrm{d}k = \int_{\mathbb{R}} f_j(x)\widehat{\varphi}(x)\,\mathrm{d}x.$$

By Plancherel (Theorem 7.7), the transforms tend in L^2 to the Plancherel transform $\widehat{f}$. Both φ and $\widehat{\varphi}$ belong to L^2. Hölder therefore passes each integral to its limit. The resulting identity is $\int \widehat{f}\varphi = \int f\widehat{\varphi} = (\mathcal{F}f)[\varphi]$. This proves agreement on every test. ∎

The differentiation rules retain the signs of Lecture 7:

$$\mathcal{F}(T') = \mathrm{i}k\mathcal{F}T, \qquad \mathcal{F}(xT) = \mathrm{i}(\mathcal{F}T)'.$$

Indeed, $(\mathcal{F}\varphi)' = \mathcal{F}(-\mathrm{i}k\varphi)$ and $\mathcal{F}(\varphi') = \mathrm{i}x\mathcal{F}\varphi$ on tests. Substitution into the two definitions gives these distributional rules.

Theorem 8.4 (Fourier table). *With the unitary kernel $(2\pi)^{-1/2}e^{-\mathrm{i}kx}$,*

T	$\mathcal{F}T$
δ	$(2\pi)^{-1/2}$
1	$\sqrt{2\pi}\,\delta$
x	$\mathrm{i}\sqrt{2\pi}\,\delta'$

The remaining three transforms are

$$\mathcal{F}\theta = \frac{\pi\delta - \mathrm{i}\,\mathrm{PV}(1/k)}{\sqrt{2\pi}}.$$

$$\mathcal{F}(\operatorname{sgn} x) = -\mathrm{i}\sqrt{2/\pi}\ \mathrm{PV}(1/k).$$

$$\mathcal{F}\big(\mathrm{PV}(1/x)\big) = -\mathrm{i}\sqrt{\pi/2}\ \operatorname{sgn} k.$$

Proof. Evaluation at zero gives $(\mathcal{F}\delta)[\varphi] = \widehat{\varphi}(0) = (2\pi)^{-1/2}\int \varphi$. The inversion evaluation Corollary 7.6 gives $(\mathcal{F}1)[\varphi] = \int \widehat{\varphi} = \sqrt{2\pi}\varphi(0)$. The multiplication rule then proves the row for x.

For $\varepsilon > 0$, integrate the decaying exponential to obtain

$$\mathcal{F}(\theta(x)e^{-\varepsilon x})(k) = \frac{(2\pi)^{-1/2}}{\varepsilon + \mathrm{i}k} = \frac{-\mathrm{i}(2\pi)^{-1/2}}{k - \mathrm{i}\varepsilon}.$$

The original functions converge to θ in $\mathcal{S}'$ by dominated convergence (Theorem 3.5) on each test. Their transforms therefore converge to $\mathcal{F}\theta$. Plemelj gives exactly the displayed row for θ. This argument determines its delta term as well as its principal value term. Using $\operatorname{sgn} = 2\theta - 1$ cancels the delta terms and gives the row for sgn. Finally, apply $\mathcal{F}$ again to that row. Reflection gives $\mathcal{F}^2(\operatorname{sgn}) = -\operatorname{sgn}$. Dividing by $-\mathrm{i}\sqrt{2/\pi}$ gives the row for $\mathrm{PV}(1/x)$. ∎

For example, the nonintegrable plane wave $e^{\mathrm{i}bx}$, with $b \in \mathbb{R}$, has transform $\sqrt{2\pi}\delta_b$. Indeed, inversion at b gives $\int e^{\mathrm{i}bx}\widehat{\varphi}(x)\,\mathrm{d}x = \sqrt{2\pi}\varphi(b)$. The frequency is concentrated at $k = b$ in this precise distributional sense.

8.5 Fundamental solutions on the line

For a constant-coefficient differential operator L, a *fundamental solution* is a distribution E satisfying $LE = \delta$. The letter E denotes that solution here and in § 8.6.

Theorem 8.5 (Two fundamental solutions). *For* $\operatorname{Re}\beta > 0$ *and* $\operatorname{Re}\eta > 0$*, respectively,*

$$(\mathrm{d}/\mathrm{d}t + \beta)E = \delta, \quad E(t) = \theta(t)e^{-\beta t}; \qquad (-\mathrm{d}^2/\mathrm{d}x^2 + \eta^2)E = \delta, \quad E(x) = \frac{e^{-\eta|x|}}{2\eta}.$$

Both functions belong to $L^1(\mathbb{R})$*. The second has transform* $\widehat{E}(k) = (2\pi)^{-1/2}/(k^2 + \eta^2)$.

Proof. For the first function and $\varphi \in \mathcal{S}$, integration by parts gives

$$E[-\varphi' + \beta\varphi] = -\int_0^\infty (e^{-\beta t}\varphi(t))'\,\mathrm{d}t = \varphi(0).$$

The boundary term at infinity vanishes since $\operatorname{Re}\beta > 0$. This solution is *retarded*: it vanishes for negative time.

The second function is continuous and solves $-E'' + \eta^2 E = 0$ off zero. Its one-sided derivatives are $E'(0+) = -1/2$ and $E'(0-) = 1/2$. For a continuous function with piecewise smooth derivative, integration by parts on both half-lines gives

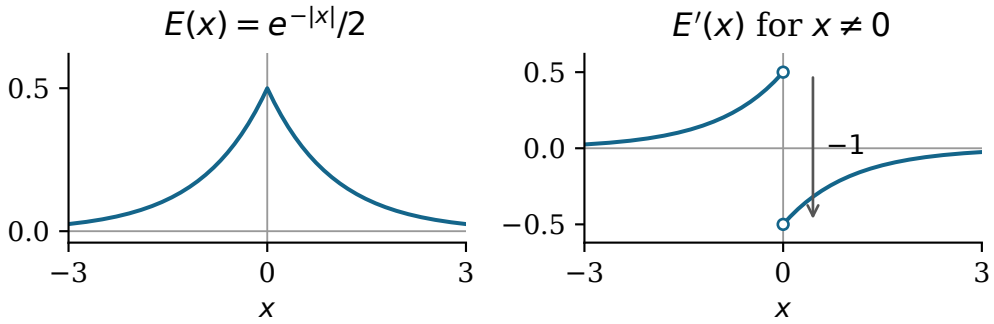
$$E''[\varphi] = \int_{x\neq 0} E''(x)\varphi(x)\,\mathrm{d}x + [E'(0+) - E'(0-)]\varphi(0).$$

The boundary terms containing $\varphi'(0)$ cancel by continuity of E. The derivative jump is -1, so $-E'' + \eta^2 E = \delta$. Exponential decay gives integrability of both solutions. Finally, direct integration on the half-lines gives

$$\widehat{E}(k) = \frac{(2\pi)^{-1/2}}{2\eta}\left(\frac{1}{\eta + \mathrm{i}k} + \frac{1}{\eta - \mathrm{i}k}\right) = \frac{(2\pi)^{-1/2}}{k^2 + \eta^2}.$$

The denominators cannot vanish for real k, since $\operatorname{Re}\eta > 0$. ∎

For $\eta = 1$, the diagram shows the continuous solution and its derivative jump. That jump supplies the point source.



The derivative has limits $1/2$ and $-1/2$ at zero. The arrow records their difference, -1.

8.6 Solving the forced equations by convolution

Theorem 8.6 (Convolution solution). *Let E be either solution in Theorem 8.5, and let L be its operator. For $f \in \mathcal{S}(\mathbb{R})$, the function $u = E * f$ is smooth and satisfies $Lu = f$.*

Proof. Since $E \in L^1$ and every derivative of f is bounded, differentiation under the integral, by Theorem 4.4, gives $u^{(j)} = E * f^{(j)}$ for every integer $j \geq 0$. The bound $|E| \sup \left|f^{(j+1)}\right|$ justifies the difference quotients. For fixed x, the function $\varphi_x(y) = f(x - y)$ belongs to $\mathcal{S}$. For $L = \mathrm{d}/\mathrm{d}x + \beta$, its distributional action gives

$$(Lu)(x) = E[-\varphi_x' + \beta\varphi_x] = (LE)[\varphi_x] = f(x),$$

since $-\varphi_x'(y) = f'(x - y)$. For $L = -\mathrm{d}^2/\mathrm{d}x^2 + \eta^2$, use instead

$$(Lu)(x) = E[-\varphi_x'' + \eta^2\varphi_x] = (LE)[\varphi_x] = f(x).$$

The coefficients have no complex conjugates, because the pairing is bilinear. ∎

For example, the first equation has the retarded solution $u(t) = \int_{-\infty}^{t} e^{-\beta(t-s)} f(s)\,\mathrm{d}s$. Only forcing at times $s \leq t$ enters its value.

For a particle of mass $m > 0$, put $H_0 = -(\hbar^2/2m)\mathrm{d}^2/\mathrm{d}x^2$ and $z = -\hbar^2\eta^2/(2m)$. Multiplying the second fundamental solution by $2m/\hbar^2$ gives

$$(H_0 - z)^{-1}(x, y) = \frac{m}{\hbar^2\eta} e^{-\eta|x-y|}, \qquad \operatorname{Re}\eta > 0.$$

This is the free resolvent kernel used in QM-II. Here the notation denotes the integral solution supplied by Theorem 8.6. For L^2 forcing it is also a bounded operator by Young's inequality (Theorem 5.6). Approximate the forcing in L^2 by Schwartz functions, using Theorem 5.8. Young's bound gives convergence of the solutions in L^2. For each Schwartz test, the test and its derivatives belong to L^2. The pairing bound (8.1) therefore passes the forced equation to the limit. For uniqueness, Fourier compatibility (Proposition 8.3) and transformation of the homogeneous equation give $(k^2 + \eta^2)\widehat{u} = 0$ as a distribution. On every bounded open interval, the function $(k^2 + \eta^2)\widehat{u}$ belongs to L^2 and has zero integral against every compactly supported smooth test. The density theorem (Theorem 5.8) then makes this function zero a.e. on the interval: approximate its complex conjugate in L^2 by such tests and use Hölder. Exhausting the line by these intervals shows $(k^2 + \eta^2)\widehat{u} = 0$ a.e. Since $k^2 + \eta^2$ has no real zero, $\widehat{u} = 0$ a.e. Plancherel (Theorem 7.7) gives $u = 0$ in L^2.

A fundamental solution alone does not impose every boundary condition: one may add a solution of the homogeneous differential equation. Distributions on higher-dimensional spaces and their partial differential equations are developed in Part II.

Summary.

- Tempered distributions are linear functionals controlled by finitely many Schwartz seminorms. Their pairing is bilinear.
- Distributional differentiation gives $\theta' = \delta$ and $(\ln|x|)' = \mathrm{PV}(1/x)$.
- Plemelj gives $1/(x \mp \mathrm{i}0) = \mathrm{PV}(1/x) \pm \mathrm{i}\pi\delta$.
- The Fourier table includes the delta term in $\widehat{\theta}$. Convolution of either displayed fundamental solution with Schwartz forcing solves its differential equation.

The same boundary-value formula returns in Lecture 12 for Cauchy integrals.

References

E. M. Stein and R. Shakarchi, *Functional Analysis*, 2011, Chapter 3, §§1.4–1.5 and Theorem 2.1. Their Fourier kernel is $e^{-2\pi \mathrm{i}x\xi}$; all constants here use our unitary convention. P. Blanchard and E. Brüning, *Mathematical Methods in Physics*, 2nd ed., 2015, §3.2 and Chapter 11, p. 163. That page writes $1/(x+\mathrm{i}0) - 1/(x-\mathrm{i}0) = -2\pi\mathrm{i}\delta$, in agreement with our convention. Further reading: R. S. Strichartz, *A Guide to Distribution Theory and Fourier Transforms*, 1994, Chapters 2 and 4.